Amazon Analytics
Business Intelligence Guide

A Complete Guide to Business Questions
and Analysis with 1 Million Rows of E-Commerce Data

Dataset: 1,000,000 Orders | 48 Columns | 200,000 Customers | 10,000 Products

Written in Simple English (A1 Level)

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1. What Does 1 Million Rows Mean for a Business?

1.1 Understanding the Size

When we say "1 million rows," it means 1,000,000 individual records. Each row is one customer order. Think about it this way: if a store gets 2,740 orders every single day for one full year, that is about 1 million orders. This is a lot of data. Real companies like Amazon handle billions of orders per year. But for learning and practice, 1 million rows is a very good size. It is big enough to show real patterns and trends, but small enough to work with on a normal computer.

In the real world, data analysts and business intelligence teams work with datasets of all sizes. Sometimes they look at a few thousand rows to answer a quick question. Sometimes they use millions of rows to find small but important patterns. With 1 million rows, you can practice both kinds of work. You can find big trends (like "which month had the most sales?") and also small patterns (like "do Prime members buy more expensive items on weekends?").

1.2 What Is Inside This Dataset?

This dataset has 48 columns (fields) of information for each order. Here is a summary of the key numbers:

Item	Value
Number of Orders	1,000,000
Number of Customers	200,000
Number of Products	10,000
Number of Columns	48
Product Categories	6 (Electronics, Computers, Smart Home, Home and Kitchen, Sports, Books)
Number of Brands	42
Time Period	Full year 2025
Geography	30 US States, 3 Regions (West, Central, East)

Table 1. Dataset Summary

1.3 Why 1 Million Rows Is Powerful

With 1 million rows, you can do many types of analysis that are not possible with smaller datasets. First, you can find statistical significance. This means you can trust your results more. For example, if you find that customers in California spend 5% more than customers in Texas, with 1 million rows you can be sure this difference is real and not just random luck. With only 100 rows, you could not trust this result.

Second, you can find segment-level patterns. This means you can look at smaller groups inside the data. For example, you can look at only "Prime members who use mobile app and buy electronics." Even this small group has thousands of rows, which is enough for good analysis. In real business, these niche segments are where companies make the most profit because they can target these groups with special offers.

Third, you can practice time-series analysis. This means looking at how things change over time. With a full year of data, you can see weekly patterns, monthly trends, and seasonal effects. For example, you can see how sales change from Monday to Sunday, or how November (Black Friday) compares to January. These time-based patterns are very important for business planning, inventory management, and marketing.

1.4 The Business Perspective

From a business point of view, 1 million orders represents a medium-sized e-commerce operation. In 2024, Amazon had over 7 billion orders worldwide. But many smaller companies and third-party sellers on Amazon handle between 100,000 and 5 million orders per year. So this dataset is very realistic for a medium-sized seller or a regional e-commerce company.

The total revenue in this dataset (if we estimate an average order of $75) would be about $75 million per year. This is the kind of revenue that a successful mid-sized online business makes. The decisions that a business analyst makes with this data have real impact: a 1% improvement in conversion rate could mean $750,000 more in revenue. A 5% reduction in return rate could save millions in logistics costs. This is why business intelligence is so valuable.

2. The 48 Columns Explained

Each row in the dataset has 48 pieces of information (columns). These columns are grouped into logical categories. Understanding what each column means is the first step to doing good analysis. Below, we explain each group of columns in simple terms.

2.1 Order and Transaction Data

These columns tell you the basic information about each order. Every row has a unique Order_ID so you can track individual orders. The Session_ID shows which shopping session the order came from. One session can have multiple orders, or a customer can visit many sessions before placing an order. Order_Date tells you exactly when the order was placed, including the hour and minute. This is useful for time-based analysis like "what hour of the day has the most orders?"

Column	What It Means
Order_ID	A unique number for each order (like AMZN-10000001)
Session_ID	The shopping session number
Order_Date	Date and time when the order was placed
Actual_Delivery_Date	When the order was delivered (empty if not delivered)
Total_Amount	The final price the customer paid (including tax and shipping)
Net_Profit	Revenue minus cost of goods and shipping
Quantity	How many units of the product were ordered
Order_Status	Delivered, Shipped, Processing, Cancelled, or Returned
Payment_Method	Credit Card, Amazon Pay, PayPal, Gift Card, Debit Card, Venmo

Table 2. Order and Transaction Columns

2.2 Product Data

These columns describe the product that was ordered. Each product has an ASIN (Amazon Standard Identification Number), which is a unique code. The Product_Category tells you which of the 6 categories the product belongs to. Product_Name includes the brand name and product type. The Unit_Price is the selling price, and COGS_Price is the cost to the seller (Cost of Goods Sold). The difference between price and COGS is the gross margin, which is a key business metric.

Column	What It Means
ASIN	Amazon product code (unique identifier)
Product_Category	Electronics, Computers, Smart Home, Home and Kitchen, Sports and Outdoors, Books
Product_Name	Full name including brand and model
Unit_Price	Selling price per unit in dollars
COGS_Price	Cost price per unit (what the seller paid)
Competitor_Price_At_Order	Price of the same product at competitors
Price_Elasticity_Score	How sensitive customers are to price changes (0.5 to 2.5)
Product_Rating	Customer rating from 3.0 to 5.0 stars

Review_Count	Number of customer reviews for the product
Buy_Box_Eligible	Yes or No - can this seller win the Buy Box?

Table 3. Product Columns

2.3 Customer Data

These columns are about the customer who placed the order. Customer_ID is a unique code for each customer. The same customer can appear in many rows because customers can place many orders. Is_Prime_Member tells you if the customer has an Amazon Prime subscription. Customer_Lifetime_Value (CLV) is the total amount of money this customer has spent across all their orders. CLV is one of the most important metrics in business because it tells you how much each customer is worth.

Column	What It Means
Customer_ID	Unique customer code (like CUST-100000)
Is_Prime_Member	1 = Prime member, 0 = not a member
Customer_Lifetime_Value	Total money spent by this customer (all orders combined)
Customer_State	US State where the order was shipped (30 states)
Customer_Region	West, Central, or East region

Table 4. Customer Columns

2.4 Marketing and Traffic Data

These columns show how the customer found the product. Traffic_Source tells you where they came from: a Google search, a TikTok ad, an Instagram influencer, Amazon internal search, a direct link, or a YouTube review. Keywords_Used shows what search words the customer typed. Ad_Campaign_ID links the order to a specific advertising campaign. This data is very important for marketing teams because it shows which channels bring the most sales.

Column	What It Means
Traffic_Source	Where the customer came from
Keywords_Used	Search words the customer typed
Ad_Campaign_ID	Advertising campaign code (if from an ad)
Device_Type	Mobile App, Desktop, or Mobile Web

Time_On_Page_Sec	How many seconds the customer spent on the product page
Click_Stream_Count	How many times the customer clicked before ordering
Cart_Abandonment_History	How many times this customer left items in cart without buying

Table 5. Marketing and Traffic Columns

2.5 Logistics and Delivery Data

These columns are about shipping and delivery. The Warehouse_ID shows which fulfillment center processed the order. The Shipping_Carrier shows which delivery company (Amazon Logistics, UPS, FedEx, DHL, USPS) handled the package. Delivery_Days_Estimated is how many days the customer was told the delivery would take. By comparing the estimated days with the actual delivery date, you can measure delivery performance. Package_Weight_kg and Package_Dimensions_cm are physical attributes that affect shipping cost and warehouse space planning.

Column	What It Means
Seller_Type	Sold by Amazon or 3rd-Party Merchant
Warehouse_ID	Fulfillment center code (like FC-LAX-1)
Shipping_Carrier	Amazon Logistics, UPS, FedEx, DHL, USPS
Delivery_Days_Estimated	Expected delivery time in days
Package_Weight_kg	Package weight in kilograms
Package_Dimensions_cm	Package size (length x width x height in cm)
Lead_Time_Days	How many days the seller needs to prepare the product
Hazmat_Status	1 if the product contains hazardous materials, 0 if not

Table 6. Logistics and Delivery Columns

2.6 Risk and Return Data

These columns help measure risk and customer satisfaction. Fraud_Score is a number between 0 and 1 that shows how likely the order is to be fraudulent (fake or stolen payment). A higher score means higher risk. Return_Probability_Score shows how likely the product is to be returned. Return_Reason tells you why a returned product was sent back (Defective, Wrong Item, Changed Mind, etc.). This information helps the business reduce losses from fraud and returns.

Column	What It Means
Return_Probability_Score	How likely the product will be returned (0.01 to 0.85)
Fraud_Score	Risk of fraud for this order (0.0 to 0.99)
Return_Reason	Why the product was returned (if returned)
Promotion_Type	Type of discount used (Lightning Deal, Coupon, etc.)
Discount_Amount	How much money was taken off the price
Tax_Amount	Tax paid on the order
Shipping_Fee	Shipping cost (free for Prime members)

Table 7. Risk and Return Columns

3. Business Questions: Sales and Revenue

The first thing any business wants to know is: how much money are we making? This section covers questions about total revenue, profit, and how sales change over time. These are the most common questions that business managers and executives ask. You can answer all of these questions using SQL queries, Power BI dashboards, or Python analysis.

3.1 Total Revenue and Profit

Q: What is the total revenue and net profit for the year?

You can add up the Total_Amount column to get total revenue. You can add up the Net_Profit column to get total profit. In SQL, you would use SUM(Total_Amount) and SUM(Net_Profit). The difference between revenue and profit tells you the total cost (COGS, shipping, taxes). A healthy e-commerce business has a net profit margin of 10-20%. If your profit margin is lower, it means costs are too high.

Q: What is the average order value (AOV)?

Average Order Value = Total Revenue / Number of Orders. In SQL, this is AVG(Total_Amount). The AOV is important because it tells you how much money each customer spends per order. If the AOV is $75 and you want to increase revenue, you can try to increase the AOV (upselling) or increase the number of orders. For example, if you recommend related products at checkout, customers might add more items to their cart.

Q: How does revenue change by month?

Group the data by month using Order_Date. For each month, calculate SUM(Total_Amount). You will see that November and December have the highest revenue because of Black Friday and holiday shopping. July also has a spike because of Prime Day. January and February are typically the slowest months. This pattern is called seasonality, and understanding it helps businesses plan their inventory and marketing.

3.2 Revenue by Category

Q: Which product category makes the most revenue?

Group the data by Product_Category and sum Total_Amount for each group. Electronics and Computers usually make the most revenue because their prices are high ($20-$1,900). But Books might sell more units even though each book is cheap ($8-$46). This is the difference between revenue (total dollars) and volume (number of items sold). Both are important: high revenue categories bring more money, while high volume categories bring more customers.

Q: Which category has the best profit margin?

Profit margin = (Unit_Price - COGS_Price) / Unit_Price. For each category, calculate the average profit margin. Books often have the highest margin (50-70%) because printing costs are low. Electronics has lower margins (25-45%) because hardware is expensive to make. A business should focus on high-margin categories when possible, but also offer low-margin categories to attract customers who might buy other things too.

3.3 Revenue by Time

Q: What day of the week has the most orders?

Extract the day of the week from Order_Date and count orders for each day. You will likely see that Saturday and Sunday have the most orders because people have more free time to shop. Monday through Friday are lower but more consistent. This information helps with staffing (customer service teams) and server capacity planning.

Q: What hour of the day do people shop the most?

Extract the hour from Order_Date and group by hour. You will see a pattern: orders are low at night (midnight to 6 AM), increase during the day, and peak in the evening (6 PM to 10 PM). This makes sense because most people shop after work. Marketing teams can use this to schedule ads and email campaigns at the best times.

Q: How do Prime Day and Black Friday affect sales?

Filter the data for specific events. Prime Day is July 15-16. Black Friday is the week of November 24-30. Compare the daily revenue during these events with normal days. You will see a massive spike - these event days can generate 5-10 times more revenue than a normal day. This shows why promotional events are so important for e-commerce.

4. Business Questions: Customer Behavior

Understanding customers is the key to growing any business. This section covers questions about who the customers are, how they behave, and how much they are worth to the business. Customer analysis helps businesses decide where to spend their marketing money and how to improve the shopping experience.

4.1 Customer Segmentation

Q: How many orders does each customer place?

Group by Customer_ID and count the orders. You will see that most customers place 1-5 orders, but some customers place 15-20 orders. These high-frequency customers are very valuable because they generate more revenue per person. The 80/20 rule often applies: 20% of customers generate 80% of revenue. Finding these top customers and keeping them happy is a key strategy.

Q: How many Prime members vs non-Prime members are there?

Count customers where Is_Prime_Member = 1 vs 0. In this dataset, about 62% of customers are Prime members. Prime members are important because they spend more, order more often, and are more loyal. Comparing the average order value and lifetime value of Prime vs non-Prime customers shows the value of the Prime program.

Q: What is the average Customer Lifetime Value (CLV)?

CLV = SUM(Total_Amount) for each customer. The average CLV tells you how much money a customer is worth over their entire relationship with the business. If the average CLV is $350 and it costs $50 to acquire a new customer, the return on investment is 7:1, which is excellent. You can also segment CLV by Prime status, region, or device type.

4.2 Device and Shopping Behavior

Q: Do mobile users buy differently than desktop users?

Compare orders by Device_Type. Mobile_App users (55% of traffic) tend to spend less time on each page (8-90 seconds) and click fewer times (2-12 clicks). Desktop users spend more time (45-480 seconds) and click more (6-35 times). This might mean desktop users research products more carefully before buying. The business should make the mobile app faster and simpler, while desktop can show more detailed information.

Q: What is the cart abandonment rate?

Cart_Abandonment_History shows how many times a customer left items in their cart without buying. If most customers have 0-1 abandoned carts, that is good. If many customers have 3-7 abandoned carts, the business has a problem. Common reasons for cart abandonment: high shipping costs, complicated checkout, or the customer found a better price elsewhere. Solutions include free shipping, simplified checkout, and retargeting ads.

4.3 Geographic Analysis

Q: Which states generate the most revenue?

Group by Customer_State and sum Total_Amount. Large states like California (CA), Texas (TX), Florida (FL), and New York (NY) will generate the most revenue simply because they have more people. But if you look at revenue per customer, smaller states might perform better. This helps decide where to open new warehouses and where to run local marketing campaigns.

Q: Is there a difference between regions?

Compare West, Central, and East regions. Look at average order value, delivery times, and return rates. If one region has much longer delivery times, the business might need a new warehouse there. If one region has higher return rates, there might be a quality issue with products shipped to that area.

5. Business Questions: Marketing and Traffic

Marketing is how a business finds new customers and keeps existing ones buying. This section covers questions about which marketing channels work best, which keywords bring the most sales, and how advertising campaigns perform. Marketing analysis helps businesses spend their advertising budget more wisely.

5.1 Traffic Source Analysis

Q: Which traffic source brings the most orders?

Group by Traffic_Source and count orders. The top sources are usually Amazon Internal Search (30%) and Google Search (28%), followed by Direct Link (15%), TikTok Ad (12%), Instagram Influencer (10%), and YouTube Review (5%). Internal search is powerful because it means customers are already on Amazon and searching for products. External sources like TikTok and Instagram bring new customers who might not have found the products otherwise.

Q: Which traffic source brings the highest-value orders?

Group by Traffic_Source and calculate AVG(Total_Amount). This is different from the total count. For example, YouTube Review might bring fewer orders but each order might have a higher value because the customer watched a detailed review and is more confident about buying. TikTok might bring many small orders. Knowing this helps decide which channels to invest in for quality vs quantity.

Q: Which keywords bring the most sales?

Filter rows where Keywords_Used is not empty, then group by Keywords_Used and sum Total_Amount. This shows which search terms are most valuable. For example, "best wireless headphones" might bring $500,000 in revenue while "cheap wireless headphones" might bring $200,000. The business can use these keywords in their SEO strategy and advertising.

5.2 Advertising Campaign Performance

Q: How do ad campaigns perform?

Filter rows where Ad_Campaign_ID is not empty. Group by Ad_Campaign_ID and calculate: number of orders, total revenue, average order value, and return rate. Compare different campaigns to see which ones are most profitable. A good campaign has high revenue, low return rate, and attracts new customers. If a campaign has high sales but many returns, it might be misleading customers with false promises.

5.3 Promotion Effectiveness

Q: Which promotion type is most effective?

Group by Promotion_Type and compare metrics. Lightning Deals (15-30% discount) generate urgency and quick sales. Coupons (5-15%) attract price-sensitive customers. Subscribe and Save (5-15%) builds recurring revenue. Prime Exclusive Discount (8-12%) rewards loyalty. Compare the revenue, profit, and customer retention for each type. A promotion that brings revenue but destroys profit margin is not sustainable.

Q: How do discounts affect profit?

Compare orders with discounts vs without discounts. Calculate the average Net_Profit for discounted orders vs full-price orders. You might find that discounted orders have lower profit per order but bring more total volume. The key is to find the right balance. Also check if discounted customers come back and buy at full price later.

6. Business Questions: Logistics and Delivery

Logistics is about getting products to customers quickly and cheaply. This section covers questions about delivery performance, warehouse efficiency, and shipping costs. Good logistics is a competitive advantage because customers expect fast and reliable delivery.

6.1 Delivery Performance

Q: What percentage of orders are delivered on time?

Compare Actual_Delivery_Date with Order_Date plus Delivery_Days_Estimated. If the actual delivery took more days than estimated, it is late. In this dataset, about 15% of orders are delivered late. A 15% late rate is common in e-commerce, but reducing it to 10% would significantly improve customer satisfaction. You can also check if certain carriers or regions have higher late rates.

Q: Which shipping carrier is the most reliable?

Group by Shipping_Carrier and calculate the late delivery percentage for each carrier. Compare Amazon Logistics, UPS, FedEx, DHL, and USPS. Also look at the average delivery time for each carrier. The fastest carrier is not always the best if they have a high late rate. The business should balance speed, reliability, and cost when choosing carriers.

6.2 Warehouse Analysis

Q: Which warehouses handle the most orders?

Group by Warehouse_ID and count orders. The distribution depends on customer locations. Warehouses in high-population regions (East and West) handle more orders. If one warehouse is overloaded, the business might need to add capacity or redistribute orders to nearby warehouses.

Q: Do regional warehouses improve delivery times?

Compare delivery times when the warehouse and customer are in the same region vs different regions. Same-region orders take 1-2 days, while cross-region orders take 3-5 days. This confirms that having warehouses close to customers is important. The business should consider opening warehouses in under-served areas where delivery times are long.

6.3 Shipping Cost Analysis

Q: How much does shipping cost the business?

Shipping_Fee shows the cost for each order. Prime members get free shipping (fee is $0). Non-Prime members pay $4.99-$7.99 per order. Calculate the total shipping revenue and compare it with estimated actual shipping costs. If the business offers free shipping to Prime members, that cost must be covered by the product margin or the Prime subscription fee.

7. Business Questions: Product Performance

Knowing which products perform well and which do not is essential for product management. This section covers questions about product ratings, reviews, returns, and brand performance. These insights help the business decide which products to promote, which to improve, and which to remove.

7.1 Product Ratings and Reviews

Q: How do ratings affect sales?

Group products by Product_Rating (or create rating ranges like 3.0-3.5, 3.5-4.0, 4.0-4.5, 4.5-5.0) and calculate the total revenue and average order quantity for each group. Products with higher ratings (4.0+) should sell more because customers trust them more. If low-rated products still sell well, it might be because they are cheap or have no competition.

Q: Do products with more reviews sell more?

Create groups by Review_Count (low: 0-100, medium: 100-1000, high: 1000+). Compare the revenue and order volume for each group. Products with more reviews generally sell more because customers read reviews before buying. If a product has low sales and few reviews, the business should encourage customers to leave reviews after purchase.

7.2 Brand Performance

Q: Which brands generate the most revenue?

Extract the brand name from Product_Name (the first word before the space) and group by brand. Calculate total revenue and profit for each brand. Top brands like Sony, Samsung, Apple, and Nike will generate the most revenue. But niche brands might have better profit margins. The business should stock more of the top-selling brands and negotiate better prices with suppliers.

7.3 Return Analysis

Q: Which categories have the highest return rate?

Filter orders with Order_Status = "Returned" and group by Product_Category. Calculate the return rate as: (Returned Orders / Total Orders) for each category. Electronics typically has the highest return rate (8-25%) because products might be defective or not meet expectations. Books have the lowest return rate (1-6%). High return categories need better quality control and more accurate product descriptions.

Q: What are the most common return reasons?

Group returned orders by Return_Reason. Common reasons include: Changed Mind (25%), Defective (20%), Not as Described (20%), Better Price Found (15%), Wrong Item Shipped (10%), and Arrived Too Late (10%). Each reason requires a different solution. "Changed Mind" might need better product photos. "Defective" needs quality control. "Arrived Too Late" needs faster shipping.

8. Business Questions: Pricing Strategy

Pricing is one of the most powerful levers in business. A small change in price can have a big impact on revenue and profit. This section covers questions about price elasticity, competitive pricing, and how the Buy Box affects sales.

8.1 Price and Elasticity

Q: What is the price elasticity for each category?

Price_Elasticity_Score tells you how sensitive customers are to price changes. A score of 2.5 means very sensitive (customers will buy much less if price increases). A score of 0.5 means not sensitive (customers will buy about the same amount even if price changes). Average the elasticity score by category. Electronics might be more elastic because customers compare prices online. Books might be less elastic because readers want specific titles.

Q: How does our price compare to competitors?

Compare Unit_Price with Competitor_Price_At_Order. Calculate the average price difference by category. If your price is consistently higher than competitors, you might lose sales. If it is lower, you might be leaving money on the table. The ideal is to match or slightly beat competitor prices while maintaining good profit margins.

8.2 Buy Box Analysis

Q: Does Buy Box eligibility affect sales?

Group by Buy_Box_Eligible (Yes vs No) and compare the order volume and revenue. The Buy Box is the "Add to Cart" button on Amazon. Only one seller wins the Buy Box for each product. If you are not eligible, customers must click "See All Buying Options" to find you, which greatly reduces your chances of making a sale. About 82% of orders come from Buy Box listings.

9. Business Questions: Risk and Fraud

Fraud and risk management protect the business from losing money to fake orders, stolen credit cards, and other types of fraud. This section covers questions about fraud patterns and how to identify high-risk orders. Effective fraud detection saves businesses millions of dollars per year.

9.1 Fraud Pattern Analysis

Q: What is the average fraud score?

Calculate the average Fraud_Score across all orders. Most legitimate orders have a low score (0.0-0.3). Suspicious orders have higher scores (0.5+). Orders with status "Cancelled" often have the highest fraud scores (0.3-0.99) because cancellation is a fraud indicator. The business should set a threshold (for example, 0.5) and manually review all orders above that threshold.

Q: What factors increase fraud risk?

Analyze the correlation between Fraud_Score and other columns. You will find that: (1) New customers have higher fraud scores, (2) Orders paid with Gift Cards have higher fraud scores, (3) High-value orders (over $500) have higher fraud scores, (4) Customers with low CLV have higher fraud scores. These factors are combined to create the fraud score. This multi-factor approach is called a "risk model" and it is standard practice in e-commerce fraud prevention.

Q: How much money is at risk from fraud?

Filter orders with Fraud_Score > 0.5 and sum their Total_Amount. This shows the total value of high-risk orders. Not all of these are actual fraud, but they represent potential losses. If 5% of high-risk orders turn out to be fraudulent, the actual loss is 5% of the total. This calculation helps the business decide how much to invest in fraud prevention systems and staff.

10. Key Performance Indicators (KPIs)

KPIs are the most important numbers that a business tracks. Every CEO, manager, and analyst looks at KPIs to understand how the business is performing. Below are the most important KPIs you can calculate from this dataset, along with how to calculate them and what they mean.

KPI	How to Calculate	What It Means
Total Revenue	SUM(Total_Amount)	Total money received from all orders
Net Profit	SUM(Net_Profit)	Revenue minus all costs
Profit Margin	Net Profit / Revenue	Percentage of revenue that is profit
Average Order Value	AVG(Total_Amount)	Average spending per order
Total Orders	COUNT(Order_ID)	Total number of orders placed
Unique Customers	COUNT(DISTINCT Customer_ID)	Number of different customers
Orders per Customer	Total Orders / Unique Customers	How often customers buy
Average CLV	AVG(Customer_Lifetime_Value)	Average total spend per customer
Prime Membership Rate	COUNT(Prime) / Total Customers	Percentage of Prime members
Delivery On-Time Rate	On-Time Orders / Delivered Orders	Percentage delivered on time
Return Rate	Returned Orders / Total Orders	Percentage of orders returned
Cancellation Rate	Cancelled Orders / Total Orders	Percentage of cancelled orders
Average Fraud Score	AVG(Fraud_Score)	Average risk level of all orders
Cart Abandonment Rate	Customers with Cart_Abandon > 0 / Total	Percentage who left cart

Table 8. Key Performance Indicators from the Dataset

These KPIs are the foundation of any business dashboard. In Power BI, you would create visual cards for each KPI at the top of your dashboard. In SQL, you would write SELECT statements to calculate each one. The important thing is not just calculating these numbers, but understanding what they mean and what actions to take. For example, if the return rate is 8%, you should investigate which categories have the highest returns and why. If the AOV is declining month over month, you should check if discounting is the cause.

11. How to Analyze: Tools and Approaches

You can analyze this dataset using many different tools. The three most common tools for business analysis are SQL, Power BI, and Python. Each tool has its strengths. Below, we explain when to use each tool and give practical tips for getting started.

11.1 Using SQL

SQL (Structured Query Language) is the best tool for asking specific questions about data. You write a query, run it, and get a precise answer. SQL is fast, efficient, and works with any size of data. Here are some common SQL patterns you will use with this dataset:

Pattern	SQL Example	Purpose
Basic Aggregation	SELECT SUM(Total_Amount) FROM orders	Calculate total revenue
Group By	SELECT Product_Category, AVG(Total_Amount) FROM orders GROUP BY Product_Category	Revenue by category
Time Filter	SELECT DATE(Order_Date) as day, COUNT(*) FROM orders GROUP BY day	Daily order count
Multiple Groups	SELECT Customer_State, Product_Category, SUM(Net_Profit) FROM orders GROUP BY Customer_State, Product_Category	Profit by state and category
Subquery	SELECT * FROM orders WHERE Fraud_Score > (SELECT AVG(Fraud_Score) FROM orders)	High-risk orders
Window Function	SELECT *, RANK() OVER (PARTITION BY Product_Category ORDER BY Net_Profit DESC) FROM orders	Top products per category

Table 9. Common SQL Patterns for This Dataset

11.2 Using Power BI

Power BI is the best tool for creating interactive dashboards and visual reports. Unlike SQL, which gives you numbers in a table, Power BI shows your data as charts, graphs, and maps. This makes it easier to see patterns and share insights with others. Here is how to build a dashboard with this dataset:

Step 1: Import the CSV file into Power BI Desktop. The 48 columns will be automatically detected with their data types.

Step 2: Create a data model. Most of your analysis will use a single table, so no complex relationships are needed.

Step 3: Add KPI cards at the top: Total Revenue, Net Profit, Average Order Value, Total Orders, and Return Rate.

Step 4: Add a line chart showing Revenue by Month to see trends over time.

Step 5: Add a bar chart showing Revenue by Product Category.

Step 6: Add a map showing Revenue by Customer_State (use the state codes for location data).

Step 7: Add a pie chart showing Order Status distribution (Delivered, Shipped, Cancelled, Returned).

Step 8: Add slicers (filters) for Is_Prime_Member, Device_Type, and Traffic_Source so users can interact with the data.

11.3 Using Python

Python is the best tool for advanced analysis and machine learning. With libraries like pandas, matplotlib, and scikit-learn, you can do things that SQL and Power BI cannot do easily. Python is especially good for: (1) Cleaning and preprocessing large datasets, (2) Building predictive models, (3) Creating custom visualizations, (4) Automating repetitive analysis tasks.

For this dataset, you could use Python to build a fraud detection model (using Fraud_Score and other features as training data), a customer segmentation model (using K-means clustering on CLV, order frequency, and category preferences), or a sales forecasting model (using time-series analysis on monthly revenue data). These advanced analyses go beyond simple queries and provide predictive insights that help businesses make proactive decisions.

12. Practical Analysis Scenarios

This chapter gives you real-world scenarios that a business analyst might face. For each scenario, we explain the business problem, which columns to use, and how to approach the analysis. These scenarios combine multiple skills and are excellent for practice.

12.1 Scenario: The CEO Wants to Know Where to Invest

Business Problem: The CEO asks: "We have $500,000 for marketing next quarter. Where should we spend it?"

Approach: First, analyze which Traffic_Source brings the highest revenue and profit. Then look at which sources bring new customers (customers with only 1 order). Also check which sources have the lowest return rates. The best channel is one that brings high revenue, new customers, and low returns. You might recommend investing 40% in Google Search (high volume), 30% in TikTok Ads (new customers), and 30% in Amazon Internal Search optimization (SEO).

12.2 Scenario: Logistics Team Needs Faster Delivery

Business Problem: The logistics manager says: "15% of orders are late. Can you help us understand why?"

Approach: First, identify which orders are late by comparing Actual_Delivery_Date with Order_Date + Delivery_Days_Estimated. Then group late orders by Shipping_Carrier, Warehouse_ID, and Customer_Region. You might find that cross-region shipments are the main problem, or that one specific carrier has a much higher late rate. Also check if holiday and weekend orders are more likely to be late. The solution might be to change carriers, add warehouses, or adjust delivery estimates.

12.3 Scenario: Product Team Wants to Reduce Returns

Business Problem: The product manager says: "Our return rate is too high. What can we do?"

Approach: Analyze returned orders by Product_Category, Return_Reason, and Product_Rating. You might find that Electronics has the highest return rate and "Defective" is the top reason. For "Not as Described" returns, check if low-rated products have higher return rates. Also compare return rates between Prime and non-Prime customers. Recommendations might include: better quality control for Electronics, more detailed product descriptions, better product photos, and a stricter review process before listing new products.

12.4 Scenario: Finance Team Wants to Improve Profit Margins

Business Problem: The CFO asks: "Our profit margin is only 12%. How can we improve it?"

Approach: First, calculate the profit margin by category, brand, and promotion type. You might find that discounted orders have very low or negative margins. Check if Lightning Deals and Coupons are eating too much profit. Also compare seller types: Amazon might have better margins than third-party sellers. Look at shipping costs for non-Prime members. Recommendations might include: reducing discount depth, increasing prices for high-elasticity products, negotiating better COGS with suppliers, and encouraging more Prime sign-ups (free shipping costs are covered by subscription fees).

13. Summary: What the Business Expects

When a business has 1 million rows of order data, the management team expects the data team to answer questions that lead to actionable decisions. Here is a summary of the key business expectations:

Business Expectation	How Data Helps
Increase Revenue	Find which products, categories, and channels generate the most revenue and grow them
Reduce Costs	Identify areas where costs are high (shipping, returns, fraud) and reduce them
Improve Customer Experience	Understand what customers want and remove friction from the buying process
Optimize Marketing Spend	Know which marketing channels bring the best return on investment
Manage Risk	Detect fraud early and prevent losses from fake orders
Improve Operations	Make logistics faster and more reliable
Grow Customer Base	Find new customer segments and increase customer lifetime value
Make Data-Driven Decisions	Replace guesses with facts based on data analysis

Table 10. Key Business Expectations and How Data Helps

The most important thing to remember is that data analysis is not just about numbers. It is about finding answers to business questions and recommending actions that improve the business. Every chart, table, and number should answer the question: "So what? What should we do about this?" If your analysis shows that November sales are 3 times higher than February, the "so what" is that the business should prepare more inventory and staff for November, and consider running promotions in February to boost slow sales.

With 1 million rows and 48 columns, this dataset gives you the raw material to practice all of these skills. Whether you use SQL, Power BI, or Python, the key is to always start with a clear business question, use the right data to answer it, and communicate your findings in a way that helps the business make better decisions. This is the core skill of a business analyst, and this dataset is your training ground.